

## Sample a group

Same prompt  $q$   
 $6 \times 7 = ?$



Rollout  
snapshot

$\pi_{\theta_{\text{old}}}$

Fixed within this batch



Sample  $G$  completions



No learned critic

## Score each response



Completions



Reward

|     |    |       |   |   |   |
|-----|----|-------|---|---|---|
| ... | 42 | <eos> | → | 1 | ✓ |
| ... | 17 | <eos> | → | 0 | ✗ |
| ... | 42 | <eos> | → | 1 | ✓ |
| ... | 0  | <eos> | → | 0 | ✗ |

Outcome score for each completion

Verify the final answer

## Relative advantages

Center and scale per group

$$\hat{A}_{i,t} = \hat{A}_i = \frac{r_i - \mu_r}{\sigma_r + \delta}$$

Group mean reward: 0.5

+   -   +   -

Broadcast a positive advantage

$t_1 : +A$     $t_2 : +A$    ... :  $+A$

$\delta > 0$  stabilizes the denominator

## Optimize the policy



Current policy

$\pi_{\theta}$  Trainable

Clipped policy gradient



Reference

$\pi_{\text{ref}}$  Held fixed

KL anchor; distinct from rollout policy



Old policy defines the importance ratio  
Reference controls KL drift



Group objective: reward improvement with controlled drift

$$\hat{J}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left[ \min \left( w_{i,t} \hat{A}_i, \text{clip}(w_{i,t}, 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) - \beta k_{i,t} \right]$$



Clipped policy update

$$w_{i,t} = \frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, o_{i,<t})}$$

Surrogate clip range

1- $\epsilon$    1   1+ $\epsilon$

Clipping shapes the surrogate, not a hard step limit.



Reference KL penalty

$$k_{i,t} = u_{i,t} - \log u_{i,t} - 1, \quad u_{i,t} = \frac{\pi_{\text{ref}}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}$$

$\beta$  weights the KL anchor; rewards stay separate.

Equal rewards: no relative reward signal.

Refresh rollout snapshot; sample the next group.